

[Pranesh S A 251801234-2025-AIDS](#) ▾**P2****Started on** Thursday, 24 September 2026, 9:30 AM**State** Finished**Completed on** Thursday, 24 September 2026, 9:34 AM**Time taken** 4 mins 12 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

**Question 1** | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,3]

the subsequence is [-1,2,2,2,3]

Output:6

**Answer:** (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      int arr[100];
8      int dp[100];
9
10     int i = 0;
11     while (i < n) {
12         scanf("%d", &arr[i]);
13         i++;
14     }
15
16     i = 0;
17     while (i < n) {
18         dp[i] = 1;
19         int j = 0;
20         while (j < i)
21         {
22             if (arr[i] >= arr[j]) {
23                 if (dp[j] + 1 > dp[i]) {
24                     dp[i] = dp[j] + 1;
25                 }
26             }
27             j++;
28         }
29         i++;
30     }
31
32     int max = 0;
33     i = 0;
34     while (i < n)
35     {
36         if (dp[i] > max) {
37             max = dp[i];
38         }
39         i++;
40     }
41     printf("%d\n", max);
42     return 0;
43 }
```

|   | Input                   | Expected | Got |   |
|---|-------------------------|----------|-----|---|
| ✓ | 9<br>-1 3 4 5 2 2 2 2 3 | 6        | 6   | ✓ |
| ✓ | 7<br>1 2 2 4 5 7 6      | 6        | 6   | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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